```
shreyas@Grigori:~$ nano command_executor.c
shreyas@Grigori:~$ gcc command_executor.c -o command_executor
shreyas@Grigori:~$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 780
Child PID  : 783

Child Process
Child PID : 783
Parent PID : 780

Executing command: ls
 DIRECTORY...   OSSP               command_executor.c   fork_example.c   task1.c   week4_t1     week4_t2.c
 File1.txt     '[OSLAB]...'        file1.c              hello.c          task2     week4_t1.c
 File2.txt      command_executor   file1.txt            task1            task2.c   week4_t2

Child process completed.
shreyas@Grigori:~$ ./command_executor
Enter a Linux command: pwd

Parent Process
Parent PID : 786

Child PID  : 787
Child Process
Child PID : 787
Parent PID : 786
```

```
Parent Process
Parent PID : 780
Child PID  : 783

Child Process
Child PID : 783
Parent PID : 780

Executing command: ls
 DIRECTORY...   OSSP               command_executor.c   fork_example.c   task1.c   week4_t1     week4_t2.c
 File1.txt     '[OSLAB]...'        file1.c              hello.c          task2     week4_t1.c
 File2.txt      command_executor   file1.txt            task1            task2.c   week4_t2

Child process completed.
shreyas@Grigori:~$ ./command_executor
Enter a Linux command: pwd

Parent Process
Parent PID : 786

Child PID  : 787
Child Process
Child PID : 787
Parent PID : 786

Executing command: pwd
/home/shreyas

Child process completed.
shreyas@Grigori:~$
```

Task 2: Investigation of Hardware Resources and Operating System Services

Introduction

In this experiment, Linux terminal commands were used to study different hardware resources managed by the operating system. Commands such as uname -a, lscpu, ps, and top were executed to view system information, processor details, running processes, and system performance. This experiment helped in understanding how the operating system provides a simple way to interact with the computer hardware.

Command: uname -a

Output:

```
unameshreyas@Grigori:~$ uname -a
Linux Grigori 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64
 GNU/Linux
shreyas@Grigori:~$
```

Explanation:

The uname -a command displays information about the operating system. From the output, it is observed that the system is running Ubuntu Linux on WSL2 with a 64-bit architecture. It also displays the kernel version and system information, which helps identify the operating system environment.

Command: lscpu

Output:

```
shreyas@Grigori:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          48 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   16
  On-line CPU(s) list:    0-15
Vendor ID:                AuthenticAMD
  Model name:             AMD Ryzen 7 8845HS w/ Radeon 780M Graphics
    CPU family:           25
    Model:                117
    Thread(s) per core:   2
    Core(s) per socket:   8
    Socket(s):            1
    Stepping:             2
    BogoMIPS:             7585.85
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
                          e sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology
                           tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse
                          4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_leg
                          acy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmmc
                          all fsgsbase bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma c
                          lflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves avx512_bf16
                           clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausef
                          ilter pfthreshold avx512vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bita
                          lg avx512_vpopcntdq rdpid fsrm
Virtualization features:
  Virtualization:         AMD-V
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    256 KiB (8 instances)
  L1i:                    256 KiB (8 instances)
  L2:                     8 MiB (8 instances)
  L3:                     16 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-15
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Not affected
  Retbleed:               Not affected
  Spec rstack overflow:   Vulnerable: Safe RET, no microcode
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSB-eIBR
                          S Not affected; BHI Not affected
  Srbds:                  Not affected
  Tsa:                    Mitigation; Clear CPU buffers
  Tsx async abort:        Not affected
  Vmscape:                Not affected
shreyas@Grigori:~$ |
```

Explanation:

The lscpu command displays detailed information about the processor. The output shows that the system uses a 13th Gen Intel Core i7-13700HX processor with 24 logical CPUs. It also provides information about CPU architecture, cache memory, virtualization support, and processor features.

Command: ps

Output:

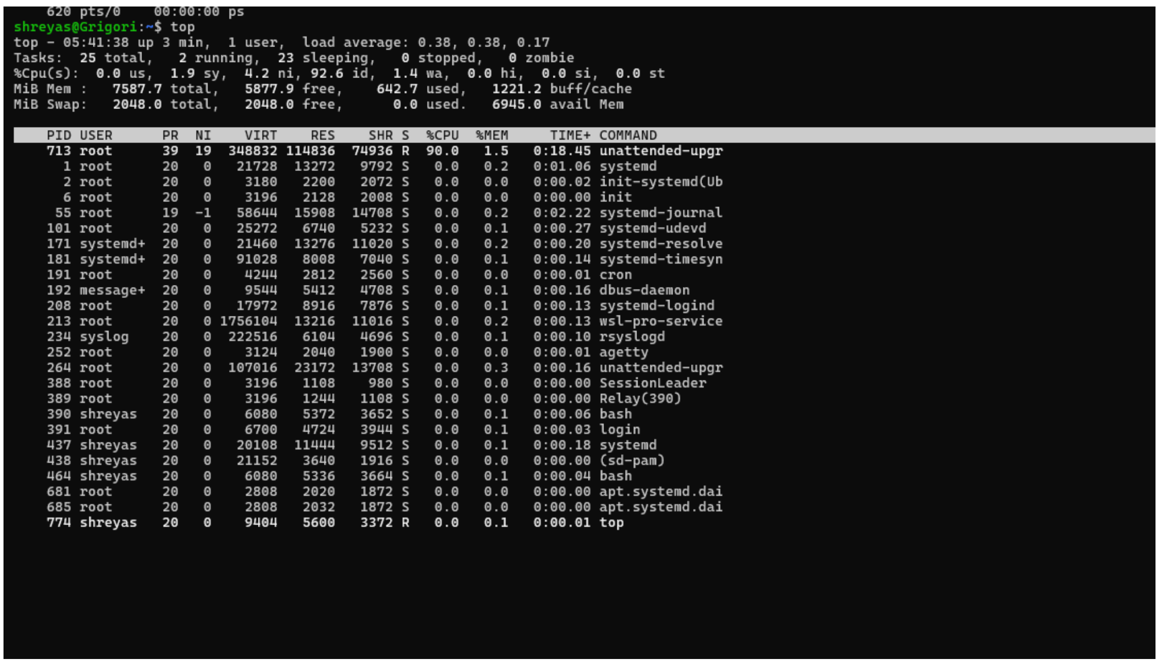
```
shreyas@Grigori:~$ ps
    PID TTY          TIME CMD
    390 pts/0    00:00:00 bash
    620 pts/0    00:00:00 ps
shreyas@Grigori:~$ |
```

Explanation:

The ps command displays the processes currently running in the terminal session. In the output, the bash shell and the ps command itself are listed along with their Process IDs (PIDs).

Command: top

Output:



```
    620 pts/0    00:00:00 ps
shreyas@Grigori:~$ top
top - 05:41:38 up 3 min,  1 user,  load average: 0.38, 0.38, 0.17
Tasks:  25 total,   2 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  1.9 sy,  4.2 ni, 92.6 id,  1.4 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7587.7 total,   5877.9 free,    642.7 used,   1221.2 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   6945.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    713 root      39  19  348832 114836  74936 R  90.0   1.5   0:18.45 unattended-upgr
      1 root      20   0   21728  13272   9792 S   0.0   0.2   0:01.06 systemd
      2 root      20   0    3180   2200   2072 S   0.0   0.0   0:00.02 init-systemd(Ub
      6 root      20   0    3196   2128   2008 S   0.0   0.0   0:00.00 init
     55 root      19  -1   58644  15908  14708 S   0.0   0.2   0:02.22 systemd-journal
    101 root      20   0   25272   6740   5232 S   0.0   0.1   0:00.27 systemd-udevd
    171 systemd+  20   0   21460  13276  11020 S   0.0   0.2   0:00.20 systemd-resolve
    181 systemd+  20   0   91028   8008   7040 S   0.0   0.1   0:00.14 systemd-timesyn
    191 root      20   0    4244   2812   2560 S   0.0   0.0   0:00.01 cron
    192 message+  20   0    9544   5412   4708 S   0.0   0.1   0:00.16 dbus-daemon
    208 root      20   0   17972   8916   7876 S   0.0   0.1   0:00.13 systemd-logind
    213 root      20   0 1756104  13216  11016 S   0.0   0.2   0:00.13 wsl-pro-service
    234 syslog    20   0  222516   6104   4696 S   0.0   0.1   0:00.10 rsyslogd
    252 root      20   0    3124   2040   1900 S   0.0   0.0   0:00.01 agetty
    264 root      20   0  107016  23172  13708 S   0.0   0.3   0:00.16 unattended-upgr
    388 root      20   0    3196   1108    980 S   0.0   0.0   0:00.00 SessionLeader
    389 root      20   0    3196   1244   1108 S   0.0   0.0   0:00.00 Relay(390)
    390 shreyas   20   0    6080   5372   3652 S   0.0   0.1   0:00.06 bash
    391 root      20   0    6700   4724   3944 S   0.0   0.1   0:00.03 login
    437 shreyas   20   0   20108  11444   9512 S   0.0   0.1   0:00.18 systemd
    438 shreyas   20   0   21152   3640   1916 S   0.0   0.0   0:00.00 (sd-pam)
    464 shreyas   20   0    6080   5336   3664 S   0.0   0.1   0:00.04 bash
    681 root      20   0    2808   2020   1872 S   0.0   0.0   0:00.00 apt.systemd.dai
    685 root      20   0    2808   2032   1872 S   0.0   0.0   0:00.00 apt.systemd.dai
    774 shreyas   20   0    9404   5600   3372 R   0.0   0.1   0:00.01 top
```

Explanation:

The top command displays the current status of the system in real time. It shows CPU usage, memory usage, running tasks, system load, and the processes currently executing.

Operating System Abstraction

CPU Abstraction

The operating system manages the CPU by deciding which process should run and for how long. This allows multiple programs to run one after another so quickly that they appear to run at the same time.

Memory Abstraction

The operating system allocates memory to each program whenever it is needed. Every process gets its own memory space, which helps prevent programs from interfering with each other.

Storage Abstraction

The operating system organizes data into files and folders. Users can easily access, create, or delete files without worrying about where the data is physically stored on the disk.

I/O Device Abstraction

The operating system communicates with hardware devices through device drivers. This allows applications to use devices like the keyboard, mouse, monitor, printer, and storage devices without directly interacting with the hardware.

Conclusion

By executing uname -a, lscpu, ps, and top, it was possible to study the operating system, processor details, running processes, and system performance. The experiment also showed how the operating system manages CPU, memory, storage, and input/output devices.